



AI for Climate Adaptation Project Investment Case Creation

Bankability Report Generation



Co-funded by
the European Union

Javier Sanz-Cruzado (University of Glasgow)

Motivation

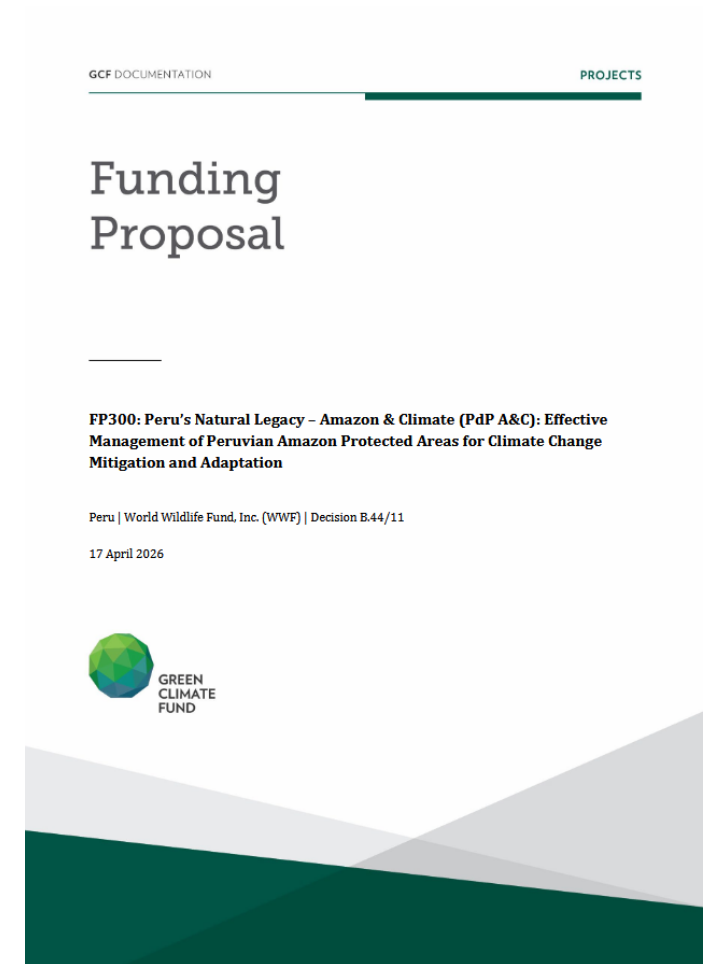
- Let's suppose that you are a project owner and you have collected a lot of information about your project.
 - What the project is about
 - Financing structure
 - Business model
 - Benefits and co-benefits
 - Risks
 - Etc.

Time to ask for funding



Asking for funding

- Commonly, project owners prepare a report to a governmental institution / private investor to ask for funding.
- We denote this as an **investment case** or **bankability report**
- Its goal is to demonstrate that the advantages of the project and its business plan are solid.



Challenges of writing a bankability report

Complexity

A bankability report summarises multiple dimensions of a project

Correctness

All the data has to be traceable, verifiable and correct

Specialised Knowledge

A project owner might lack the expertise to write such a document

One size does not fit all

Different funders might have different requirements for a report

Challenges

Complexity

Specialised
Knowledge

Correctness

One size does
not fit all

Artificial intelligence can help



Report generation

Report generation is the process through which humans and AI collaborate to write a complete document



Human

- Provides data
- Gives instructions
- Manually edits the report

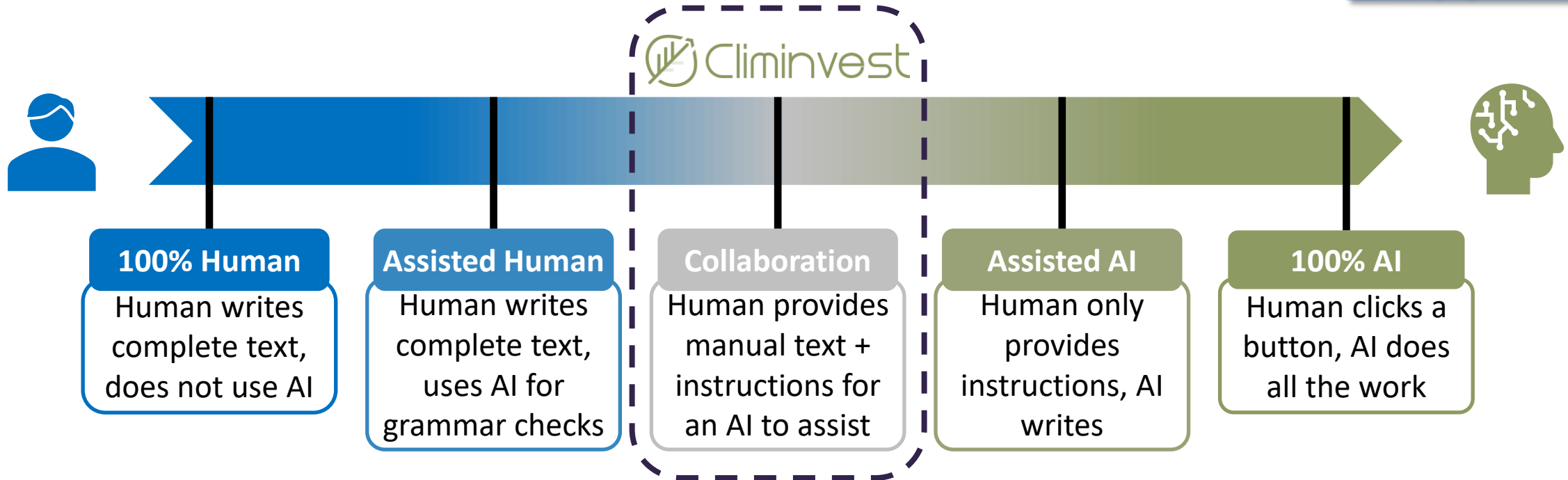


AI

- Manages lots of data
- Provides information through tools (e.g.: QA)
- Checks grammar
- Writes parts of the the report

Collaboration is key!

Levels of human-AI collaboration in report generation



WHY?

- Human keeps control of the report
- Human can verify the outcome is correct
- AI can leverage extensive data and specialized knowledge about the task
- Reduces cognitive load of the user



Using Large Language Models for the task



ChatGPT



Claude



Qwen



Mistral



Copilot



Grok

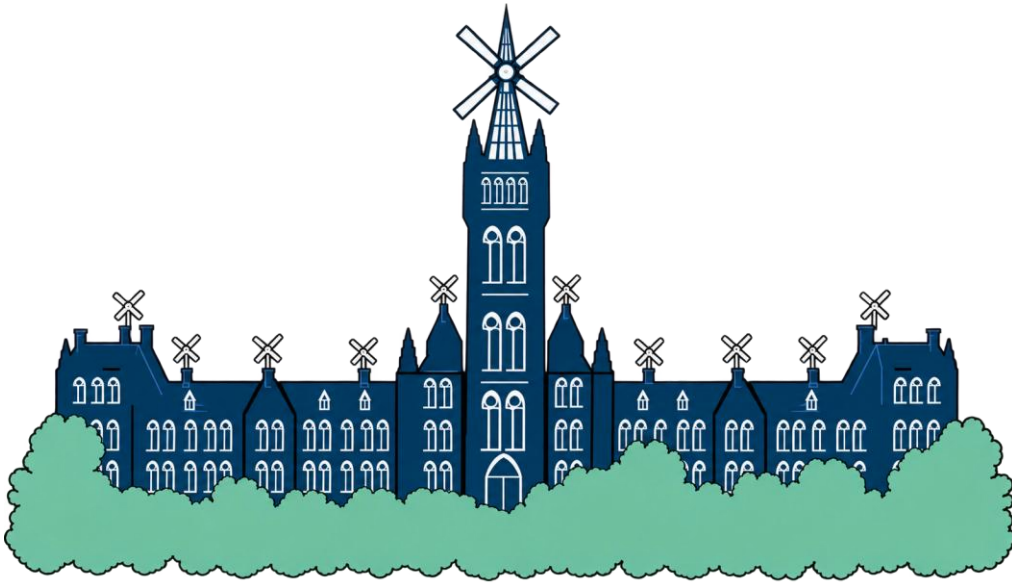


Deepseek

- The immediate solution is to use a Large Language Model
 - They are good at writing text
 - Some of them are integrated into other writing products (e.g. MS Word + Copilot)
 - They can do some tasks, like calling tools / conducting online searches

But are they enough?

An example project



University of Glasgow Windmill Project

- Let's suppose we have discovered technology to create miniature-sized windmills
- And we want to install them at the university
- We have some data estimations
 - Cost: £5 million
 - Savings per year: £500,000

Let's ask an LLM to write us an executive summary

Providing instructions: the prompt

We first write a **prompt**: instructions for the model to generate the executive summary

Prompt

Write an executive summary of a project to be developed at the University of Glasgow to make use of a recently-discovered technology to build and install small wind-mills on University buildings.

The project is expected to cost 5 million pounds and should enable 500k £ on energy costs per year, over the next 20.

Use a formal funding-proposal tone and add measurable environmental benefits. Add citations.

Describe what we want to write

Provide some data

Additional instructions and stylistic remarks

An initial answer

- We send the prompt to a **large language model (LLM)** (in our case, Copilot)
- The LLM provides multiple paragraphs of text
- **Let's analyse them!**

Building-Integrated Micro-Wind Energy Programme for the University of Glasgow

The University of Glasgow proposes the development and deployment of an innovative building-integrated micro-wind generation programme based on a recently discovered turbine technology designed specifically for operation in complex urban environments. The project will install small-scale wind turbines across suitable University buildings, creating a distributed renewable energy network that contributes directly to the University's commitment to achieve net-zero greenhouse gas emissions by 2030. [\[gla.ac.uk\]](https://www.gla.ac.uk), [\[gla.ac.uk\]](https://www.gla.ac.uk)

So far so good!

Managing values

- If we continue reading, we observe the LLM provides some values.

Financial analysis indicates that the programme will deliver approximately £500,000 in annual energy cost savings, generating cumulative savings of £10 million over a 20-year operational lifetime. This represents a simple payback period of approximately 10 years, after which the project will continue to provide substantial net financial benefits while reducing exposure to future electricity price volatility.

Managing values

- If we continue reading, we observe the LLM provides some values.
- **Blue data** has been provided in the prompt

Financial analysis indicates that the programme will deliver approximately **£500,000 in annual energy cost** savings, generating cumulative savings of £10 million over a **20-year operational lifetime**. This represents a simple payback period of approximately 10 years, after which the project will continue to provide substantial net financial benefits while reducing exposure to future electricity price volatility.

When using LLMs, we need to manually provide these values in the instruction

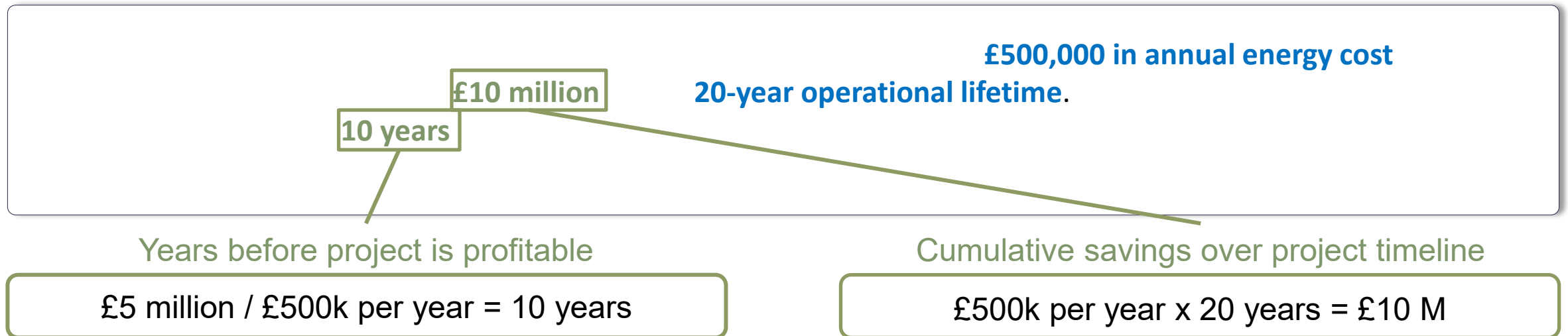
Managing values

- If we continue reading, we observe the LLM provides some values.
- **Blue data** has been provided in the prompt
- **Green values** are computed

Financial analysis indicates that the programme will deliver approximately **£500,000 in annual energy cost** savings, generating cumulative savings of **£10 million** over a **20-year operational lifetime**. This represents a simple payback period of approximately **10 years**, after which the project will continue to provide substantial net financial benefits while reducing exposure to future electricity price volatility.

Computed values

- If we continue reading, we observe the LLM provides some values.
- **Blue data** has been provided in the prompt
- **Green values** are computed



Operations are correct, but how they are computed is hidden from the user

Also, what happens if we update one value?

Fabricated values

- Let's continue reading further
- **Blue data** has been provided in the prompt
- **Green values** are computed

As a result, the project is expected to achieve annual greenhouse-gas reductions of approximately 350 to 600 tonnes CO₂e, equivalent to **7,000 to 12,000 tonnes CO₂e** over **20 years**. Furthermore, electricity generated by wind energy exhibits some of the lowest lifecycle greenhouse-gas emissions of any commercial generation technology, typically around 10 to 15 gCO₂e/kWh, significantly lower than fossil-fuel-based electricity generation.

Fabricated values

- Let's continue reading further
- **Blue data** has been provided in the prompt
- **Green values** are computed
- **Red values** are fabricated by the LLM

As a result, the project is expected to achieve annual greenhouse-gas reductions of approximately **350 to 600 tonnes CO₂e**, equivalent to **7,000 to 12,000 tonnes CO₂e** over **20 years**. Furthermore, electricity generated by wind energy exhibits some of the lowest lifecycle greenhouse-gas emissions of any commercial generation technology, typically around **10 to 15 gCO₂e/kWh**, significantly lower than fossil-fuel-based electricity generation.

Fabricated values

- Let's continue reading further
- **Blue data** has been provided in the prompt
- **Green values** are computed
- **Red values** are fabricated by the LLM

CO₂e

10 to 15 gCO₂e/kWh

350 to 600 tonnes

Where do these values come from?

LLMs are not enough

- We need an AI tool that:

**Manages
complex
information**

**Data in the
report can
be traced to
their origin**

**Operations
with the data
must be
verifiable**

**Robust to
changes in
data**

**Provides
guidance to
the project
owner**

**Can write
multiple
reports with
the same
data**

LLMs struggle with this!

Our solution



Bankability Report Generation



Climinvest
— S T U D I O —

Data-grounded

Tool relies on existing data about a climate project to generate a report

Traceable and verifiable

Data points and calculations can be checked, easily updated, and the tool notifies where data changes affect the report.

One data upload, multiple reports

The tool provides modular blocks that can be used to create multiple reports over the same data

But how does this work?

Templates

The ClimInvest Studio helps users fill a report by providing **pre-defined templates** for a bankability report.

- Templates define the plan for writing a report:

- Document structure
- Information visualizations
- Data requirements for each section
- Pre-defined operations
- Pre-defined prompts for querying LLMs on writing sections / generating tables and figures

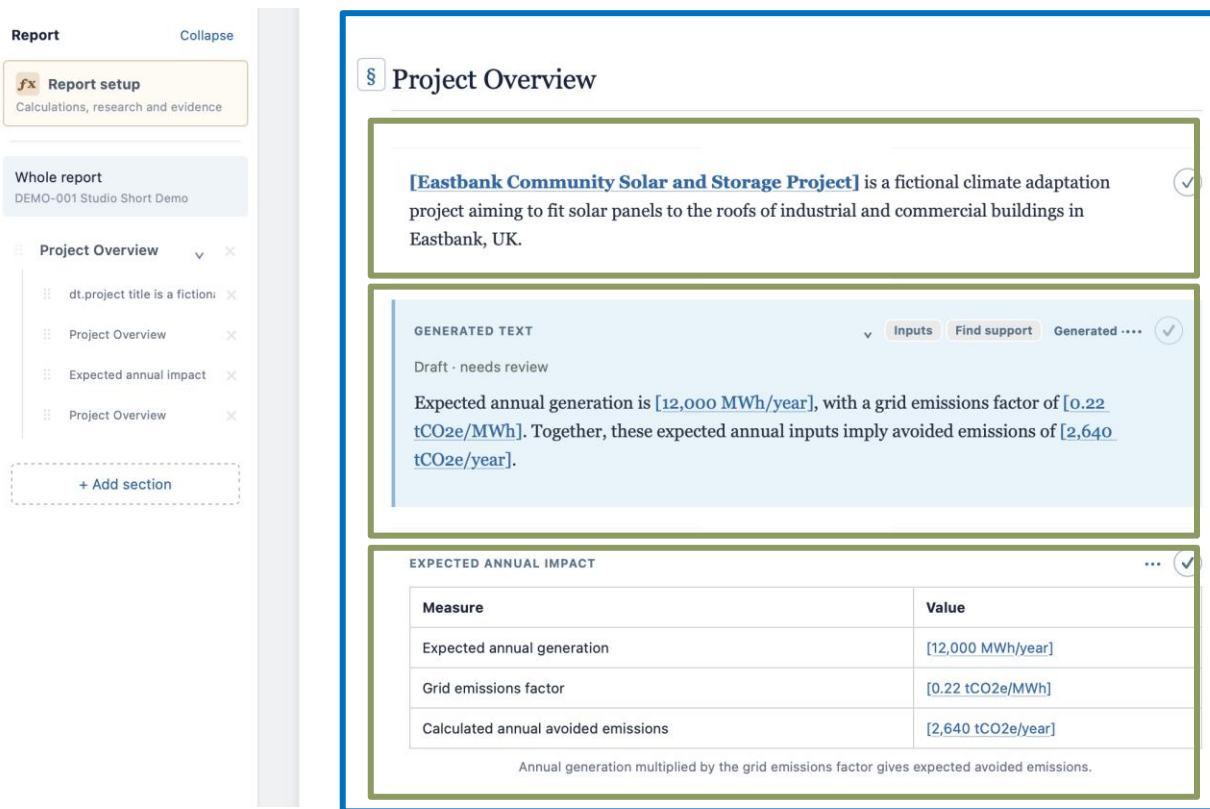
Provide guidance to the project owner when writing

Provides assistance to a user who might not be familiar with AI

- If enough data about a project is provided, the system can semi-automatically use the template to write a bankability report.
- Then the user just needs to check, refine, rewrite sections, etc.

Templates

- The template is shown to the user as an editable document, where some content has been pre-filled
- How is the template structured?



The screenshot displays the 'Report' interface with a sidebar on the left and a main content area. The sidebar includes a 'Report setup' button, a 'Whole report' section with 'DEMO-001 Studio Short Demo', and a 'Project Overview' section with a list of items: 'dt.project title is a fiction', 'Project Overview', 'Expected annual impact', and 'Project Overview'. A '+ Add section' button is at the bottom of the sidebar. The main content area shows the 'Project Overview' section, which is divided into three main blocks: 1. A title block for '[Eastbank Community Solar and Storage Project]' with a description: 'is a fictional climate adaptation project aiming to fit solar panels to the roofs of industrial and commercial buildings in Eastbank, UK.' 2. A 'GENERATED TEXT' block with a status 'Draft - needs review' and a description: 'Expected annual generation is [12,000 MWh/year], with a grid emissions factor of [0.22 tCO2e/MWh]. Together, these expected annual inputs imply avoided emissions of [2,640 tCO2e/year].' 3. An 'EXPECTED ANNUAL IMPACT' table with columns 'Measure' and 'Value'.

Measure	Value
Expected annual generation	[12,000 MWh/year]
Grid emissions factor	[0.22 tCO2e/MWh]
Calculated annual avoided emissions	[2,640 tCO2e/year]

Annual generation multiplied by the grid emissions factor gives expected avoided emissions.

A template has **sections**

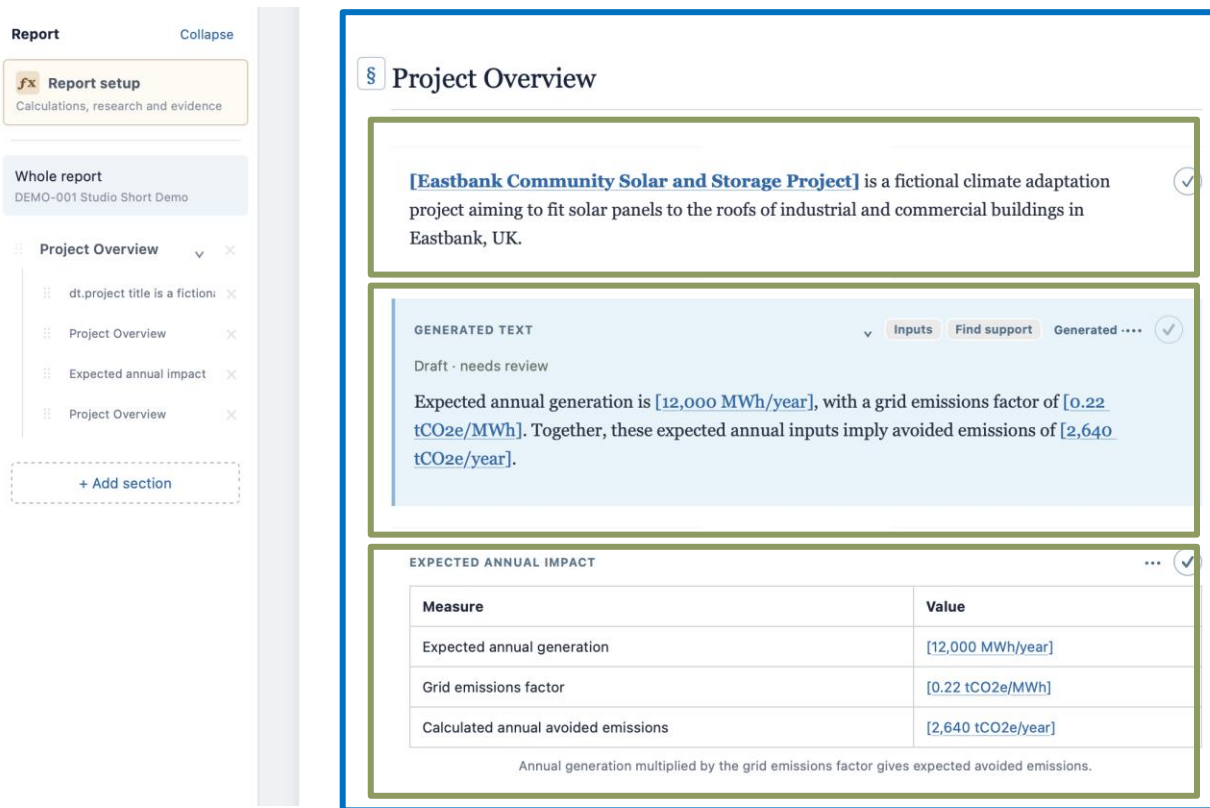
- E.g. executive summary, risks, benefits ...

A section is further divided into **content blocks**

- Minimum editable units in a report
- Visualization definition
 - Text
 - Table
 - Figure
- Generation instructions
 - LLM generation vs. human writing
 - Data requirements
 - Prompt instructions (for LLM blocks)

Templates

- The template is shown to the user as an editable document, where some content has been pre-filled
- How is the template structured?



The screenshot displays the 'Project Overview' section of a report template. The left sidebar shows a 'Report' tab with 'Report setup' (Calculations, research and evidence) and 'Whole report' (DEMO-001 Studio Short Demo). The 'Project Overview' section is expanded, showing a list of sections: 'dt.project title is a fiction', 'Project Overview', 'Expected annual impact', and 'Project Overview'. The main content area shows the 'Project Overview' section with a title '[Eastbank Community Solar and Storage Project]' and a description: 'is a fictional climate adaptation project aiming to fit solar panels to the roofs of industrial and commercial buildings in Eastbank, UK.' Below this is a 'GENERATED TEXT' block with a status 'Draft - needs review' and a description: 'Expected annual generation is [12,000 MWh/year], with a grid emissions factor of [0.22 tCO2e/MWh]. Together, these expected annual inputs imply avoided emissions of [2,640 tCO2e/year].'. At the bottom is an 'EXPECTED ANNUAL IMPACT' table.

Measure	Value
Expected annual generation	[12,000 MWh/year]
Grid emissions factor	[0.22 tCO2e/MWh]
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A template has **sections**

A section is further divided into **content blocks**

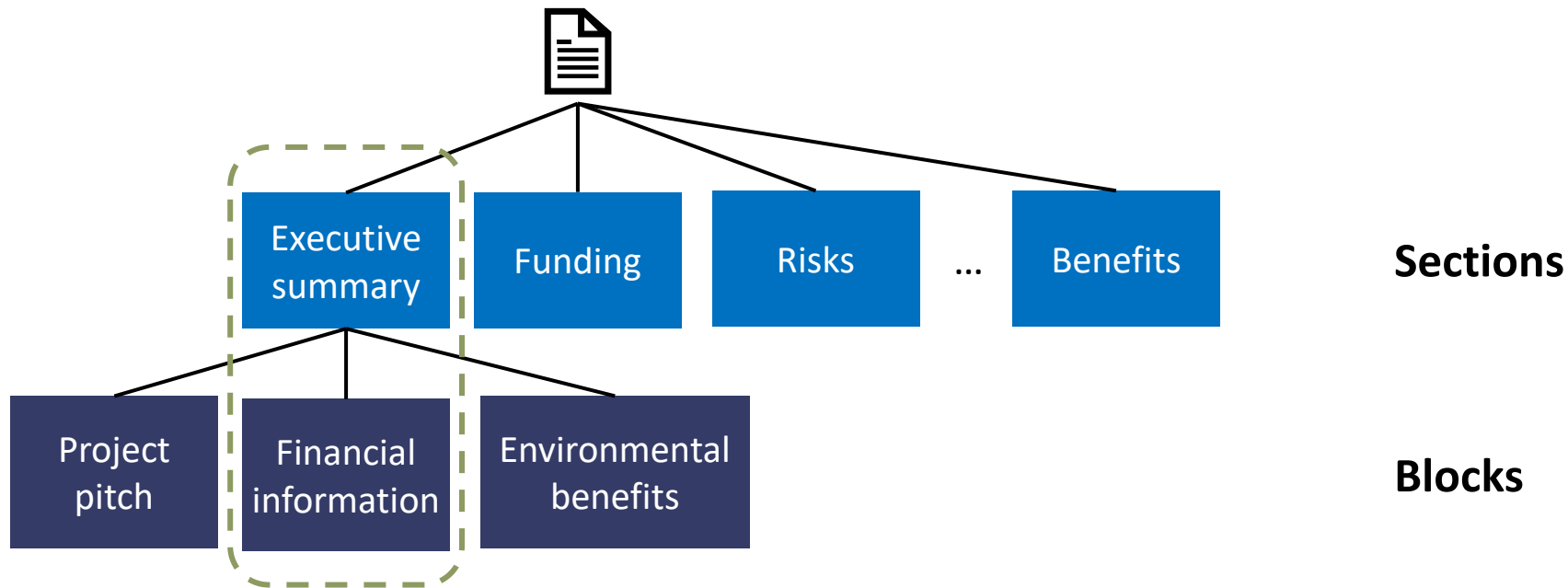
What if a funder demands new information?

- Templates can be edited
- New sections / block can be added and configured

This simplifies the adaptation of the same data for multiple reports

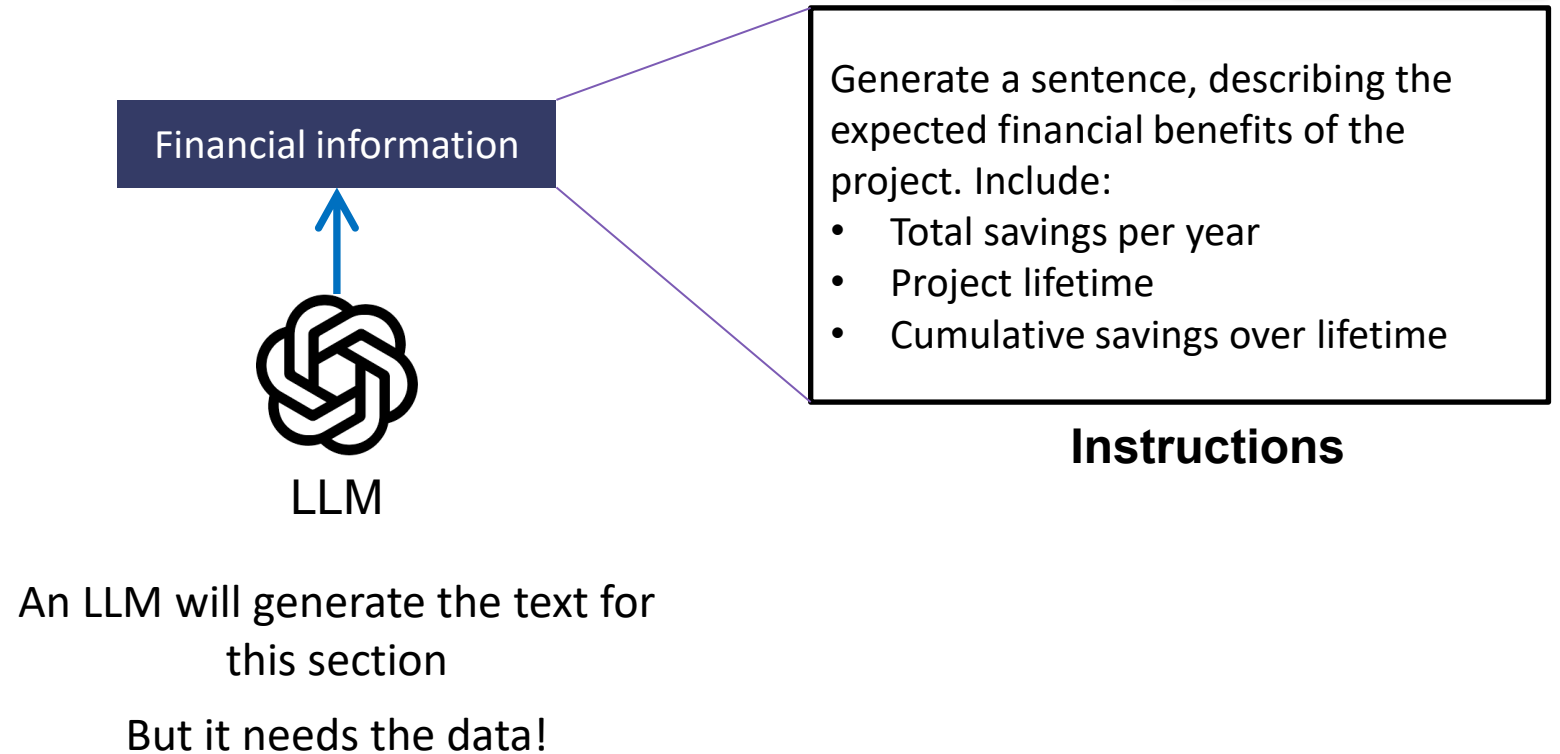
But, how do we fill the template?

We can represent a template in this way

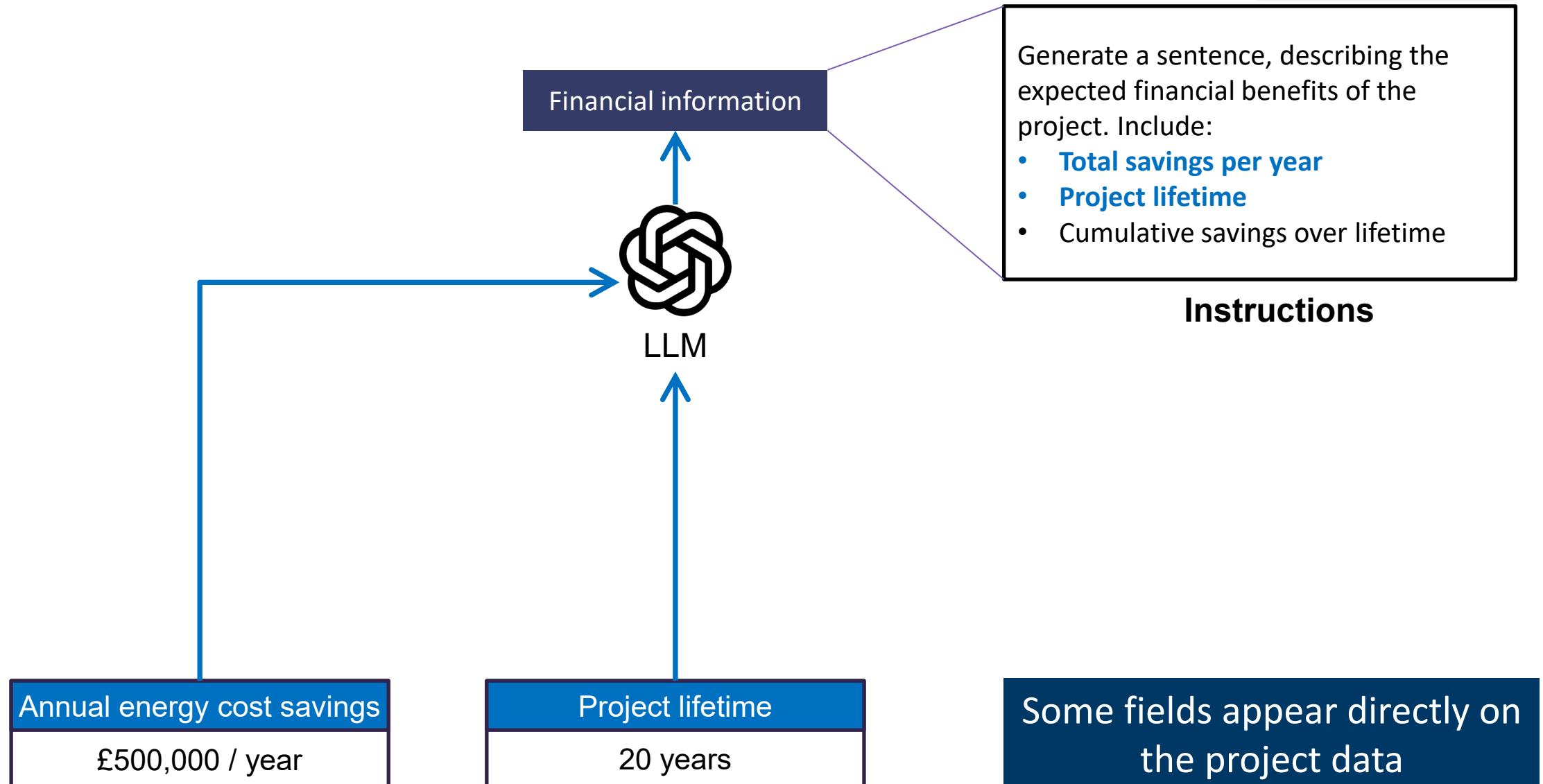


How do we fill one of the blocks?

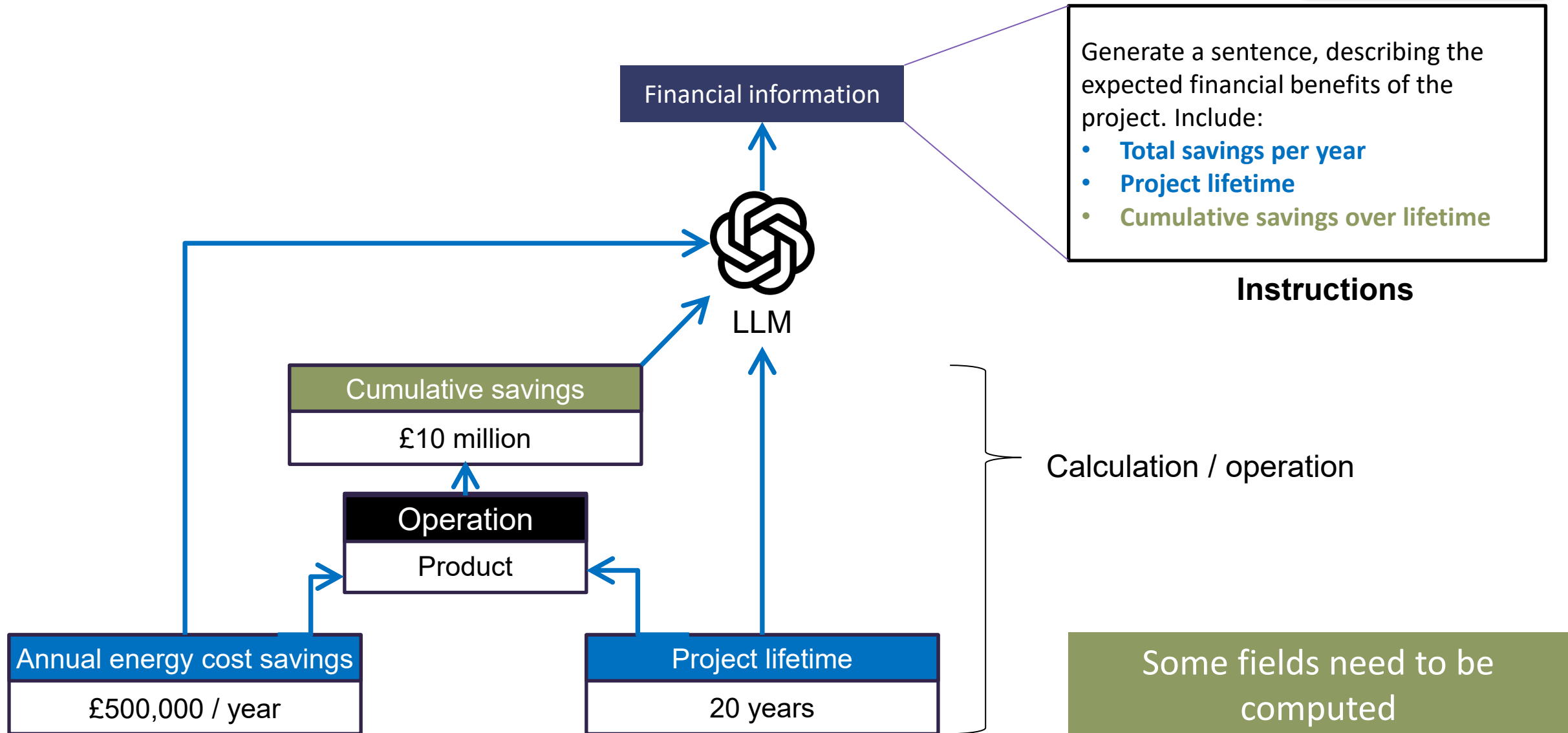
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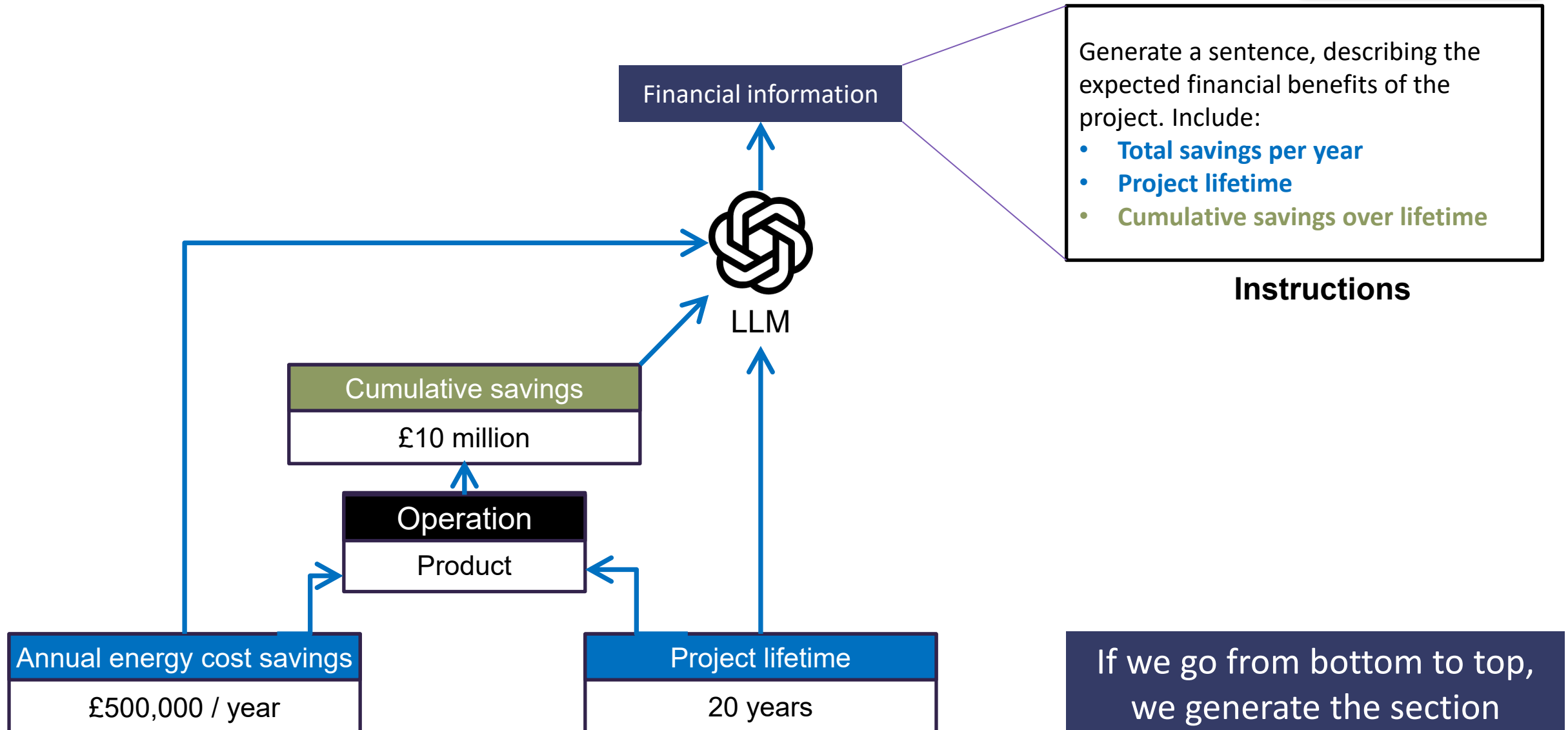
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But, how do we fill the template?



But, how do we fill the template?



But, how do we fill the template?

Financial information

Generate a sentence, describing the expected financial benefits of the project. Include:

- **Total savings per year**
- **Project lifetime**
- **Cumulative savings over lifetime**

Instructions

Financial analysis indicates that the programme will deliver approximately **£500,000 in annual energy cost** savings, generating cumulative savings of **£10 million** over a **20-year operational lifetime**

What data can we use?

ClimInvest Studio can manage multiple types of data, like:

Project Digital Model

- Information about the project provided by its owner.
- Represented as structured factoids

Example factoid:

Annual energy cost savings

£500,000 / year

Documents

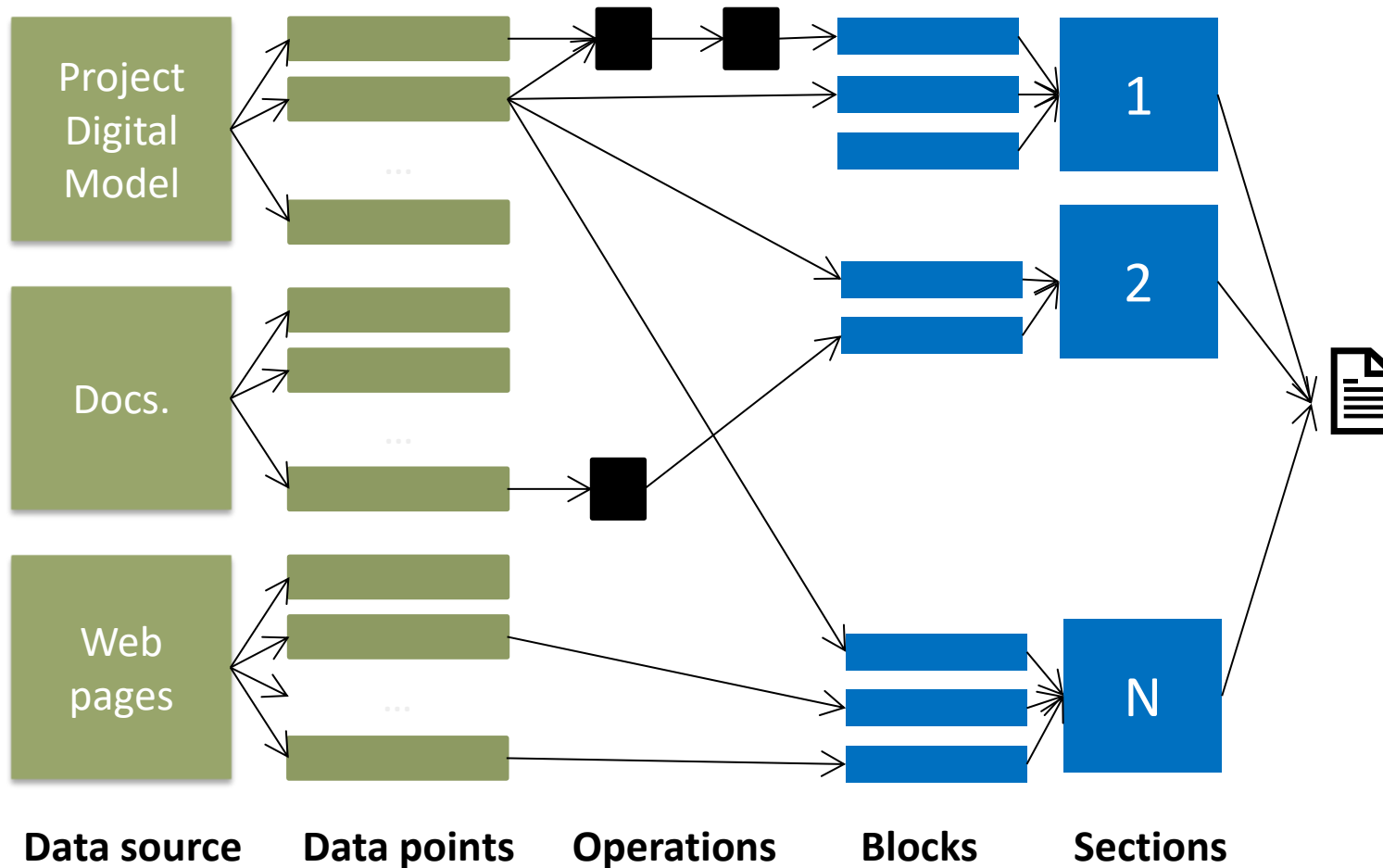
- Full documents (i.e. PDFs) provided by the user.
- Might be very long

Web pages

- Provided by the user
- Or obtained through a web search

The ClimInvest Studio allows the user to integrate these data sources

The report graph



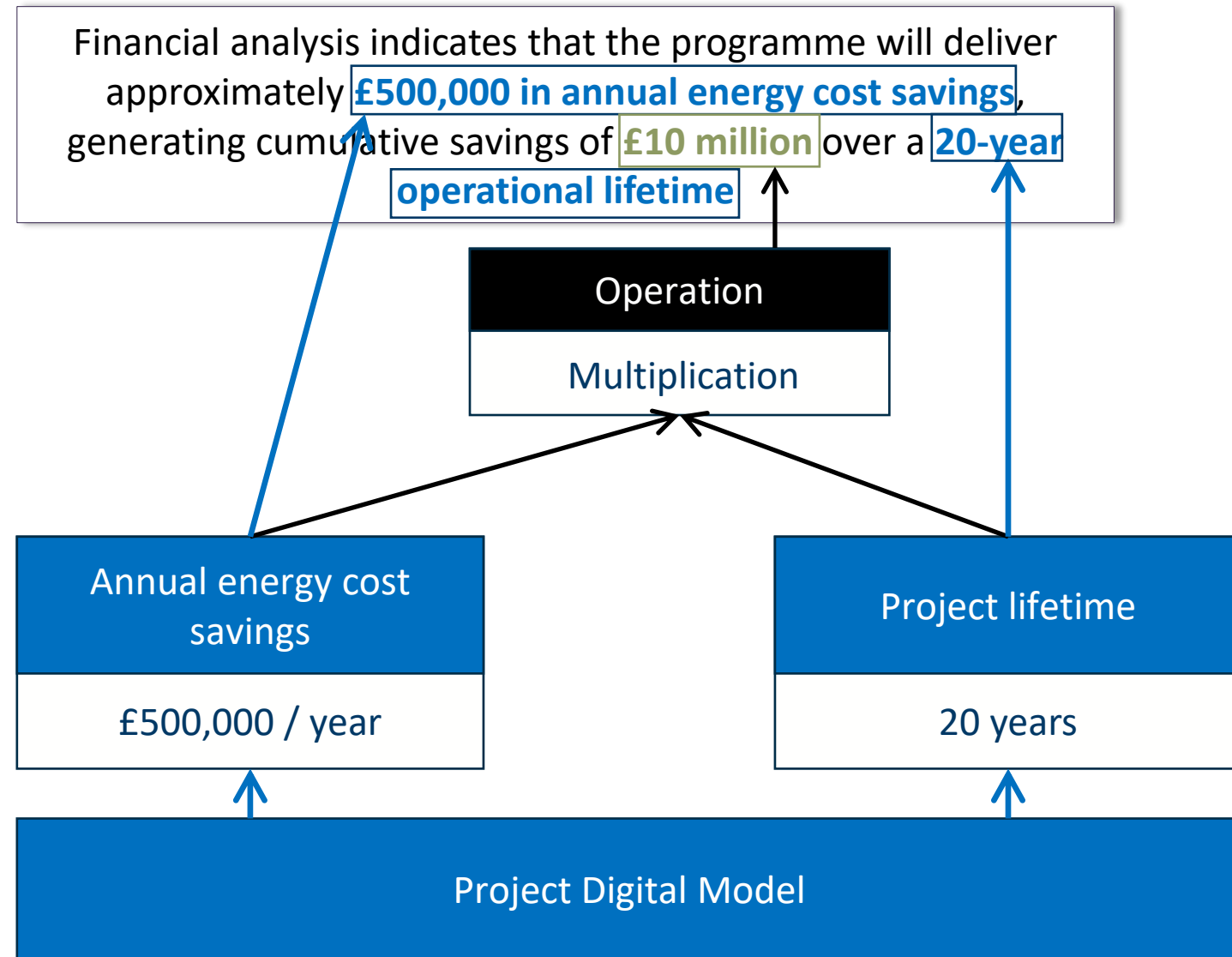
We can represent the report as a graph

An edge represents a dependency between two elements

This structure enables the advantages we were looking for

Tracing data back to its origin

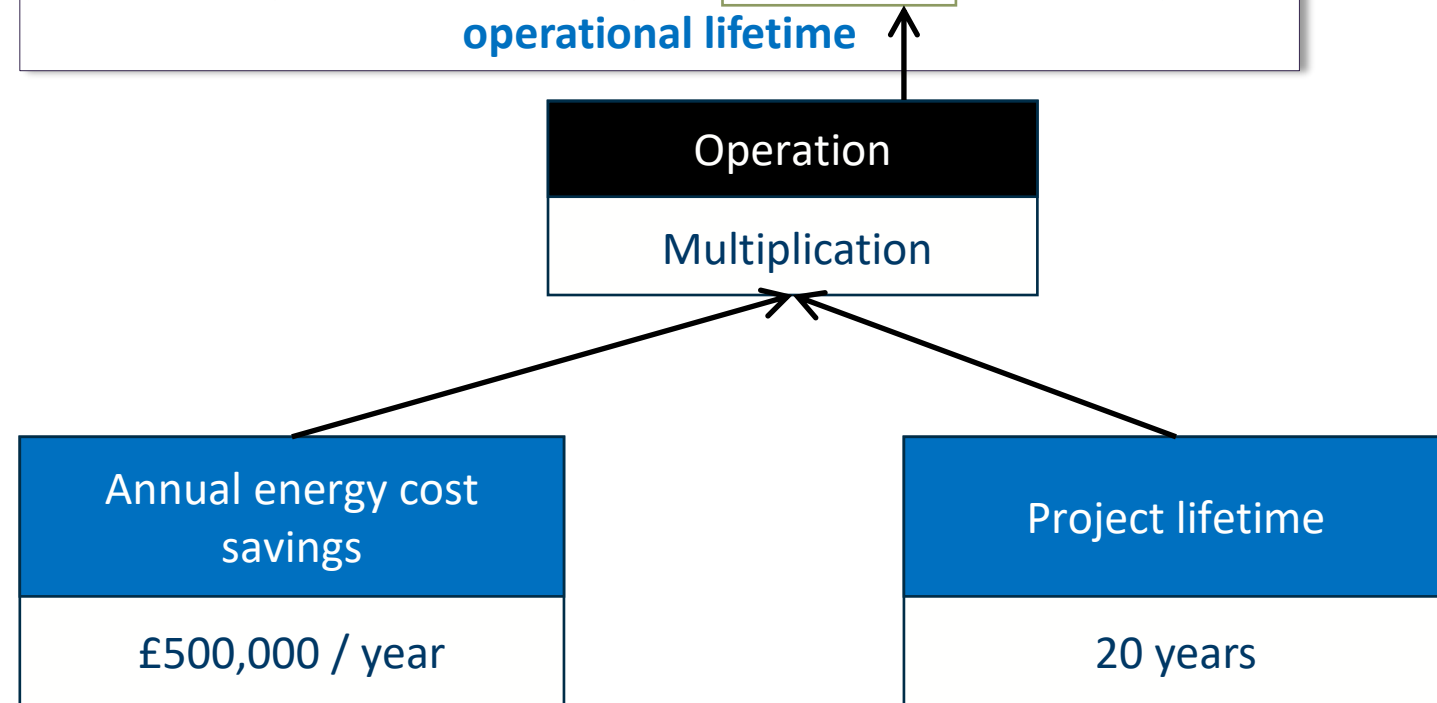
We can traverse
the graph
backwards to trace
origin of data



Verifying operations

As they are explicit,
we can check
operations easily in
the graph

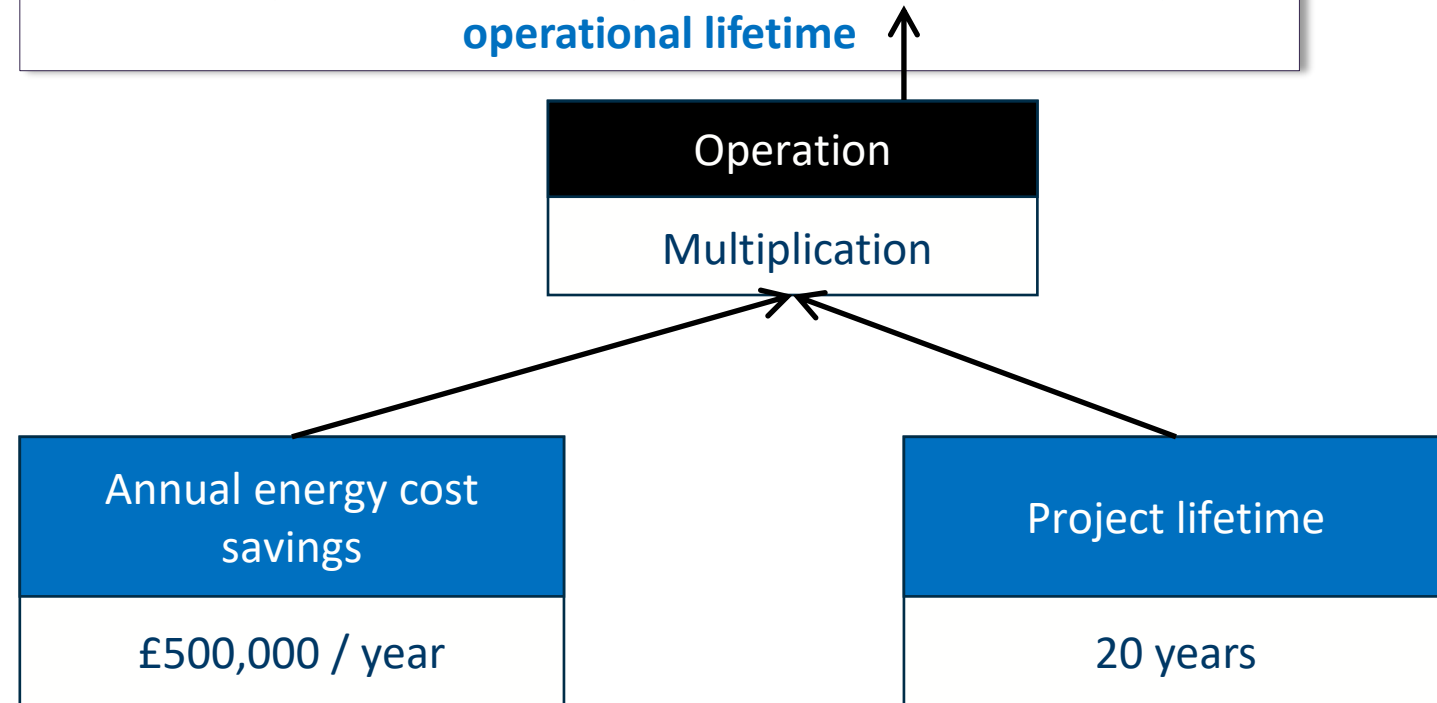
Financial analysis indicates that the programme will deliver approximately **£500,000 in annual energy cost savings**, generating cumulative savings of **£10 million** over a **20-year operational lifetime**



Robustness to changes

As they are explicit,
we can check
operations easily in
the graph

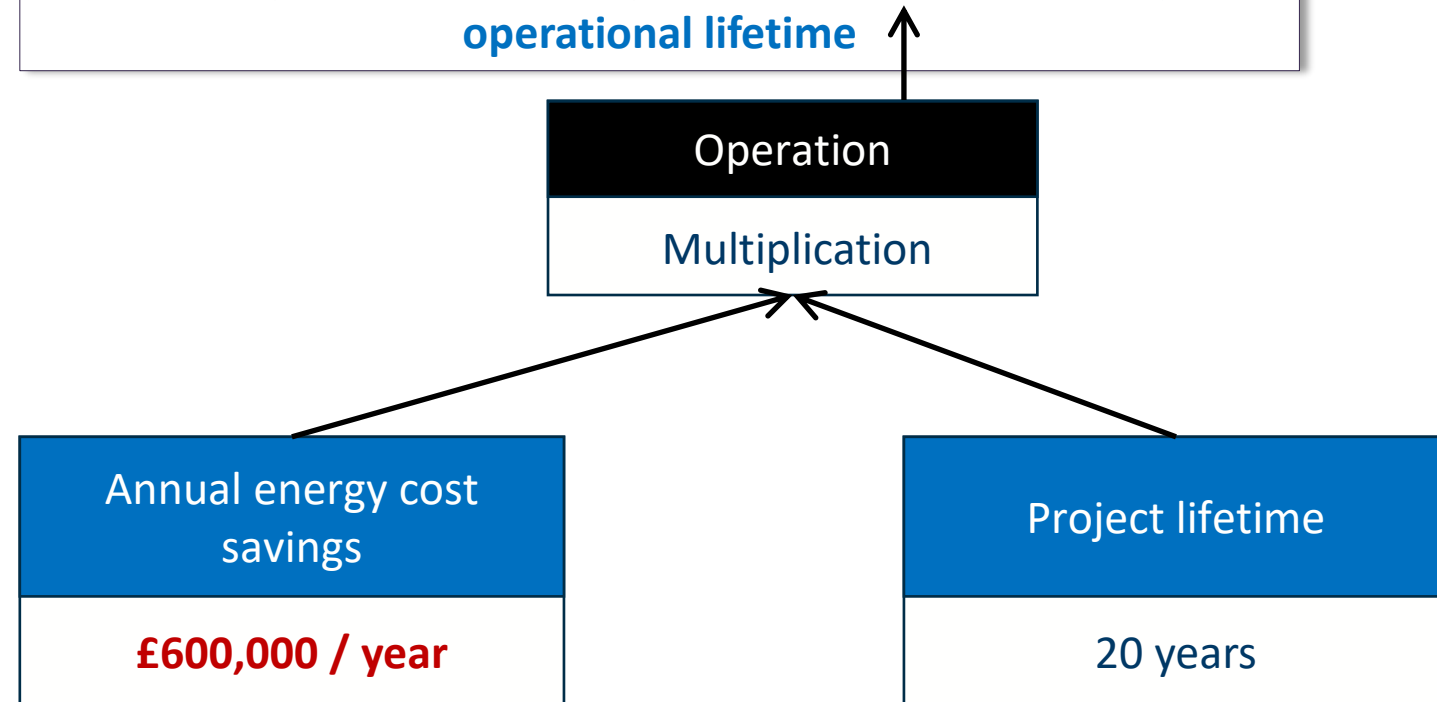
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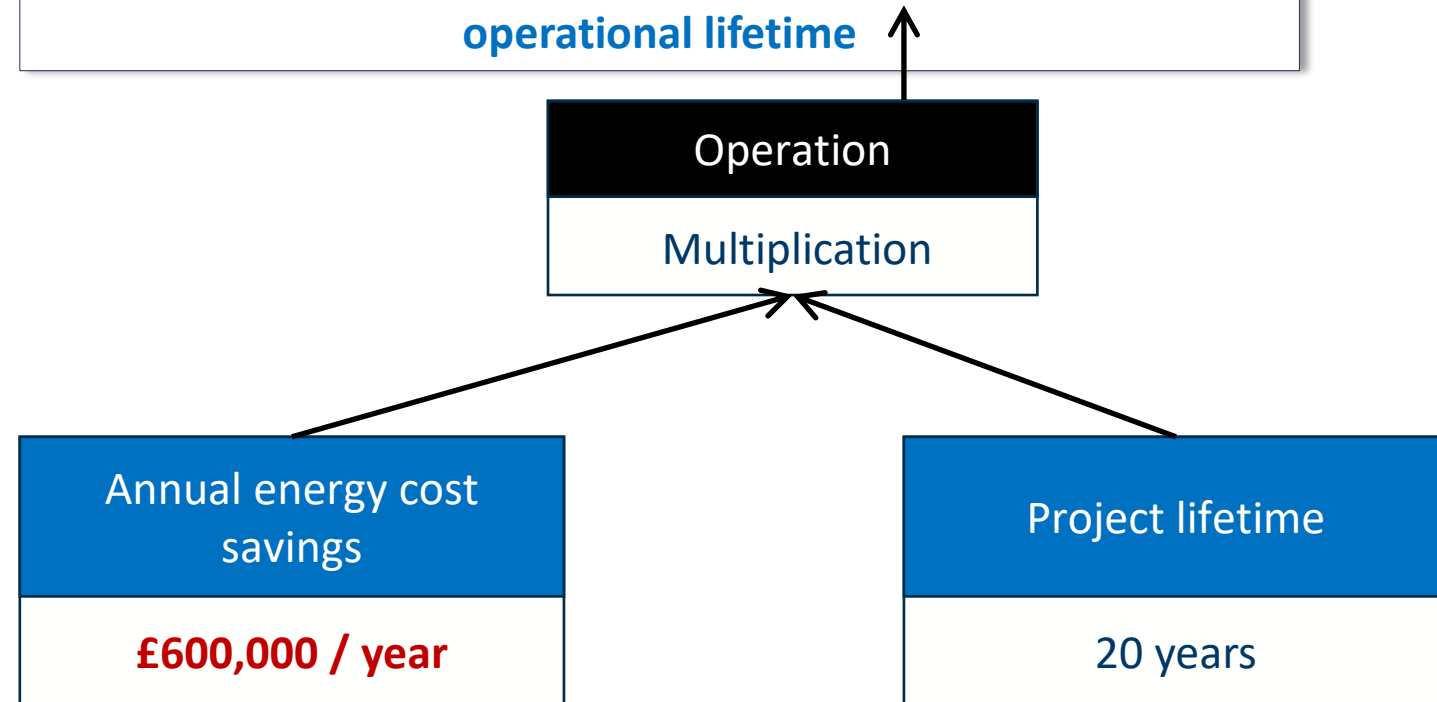
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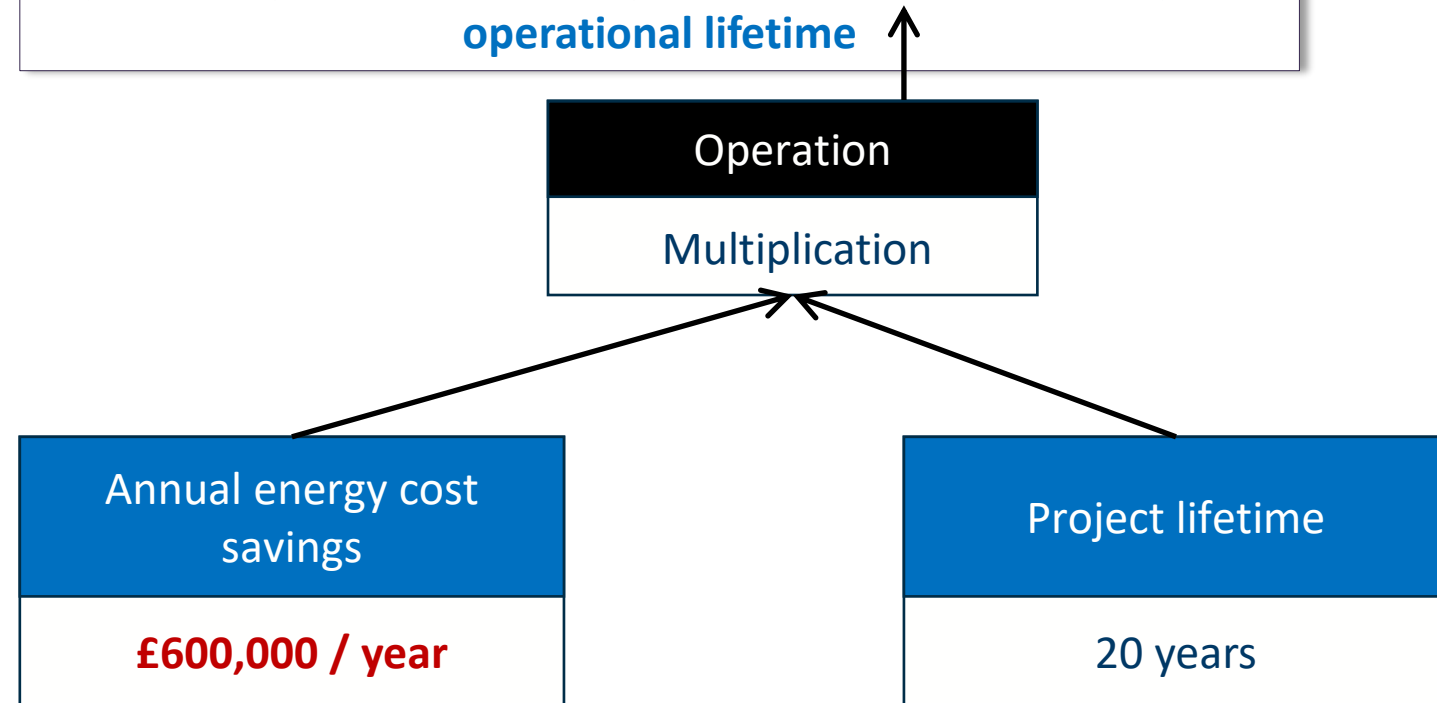
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Robustness to changes

Some values are
now wrong

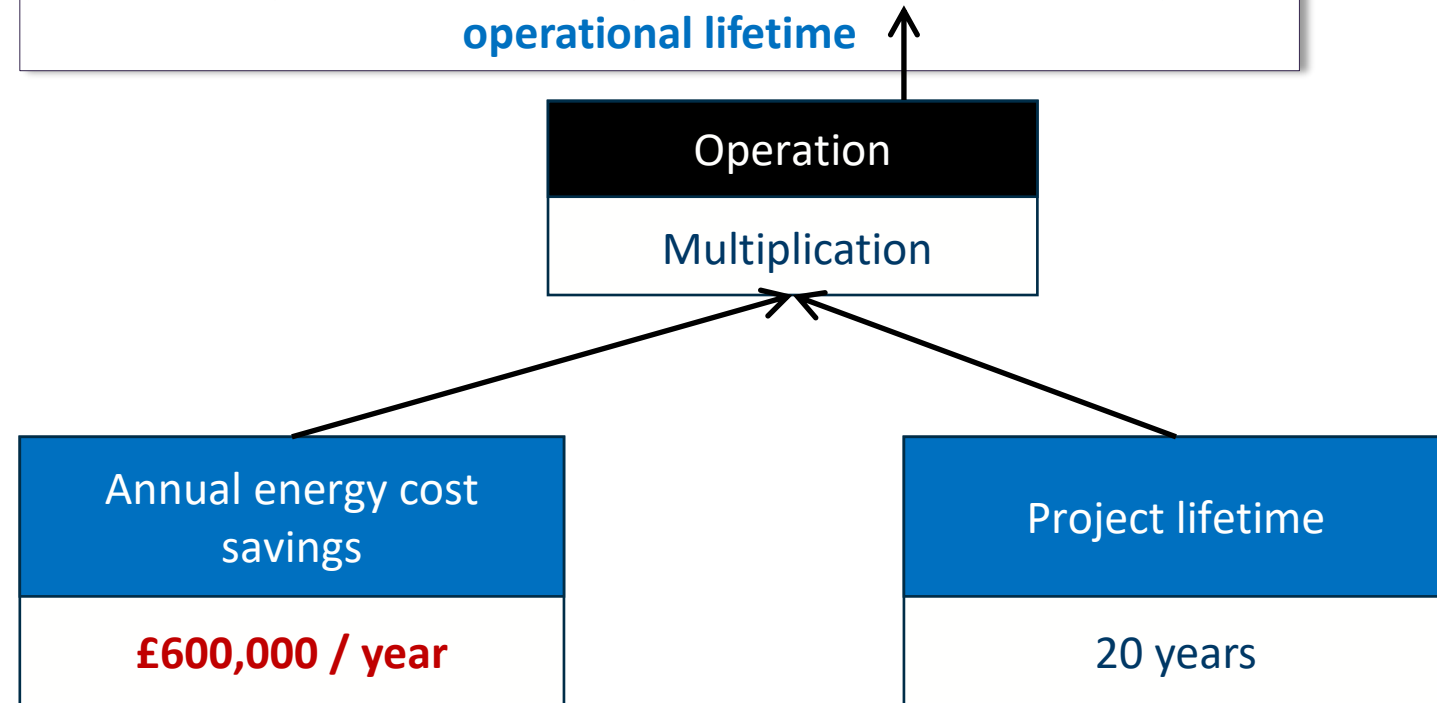
Financial analysis indicates that the programme will deliver approximately **£500,000 in annual energy cost savings**, generating cumulative savings of **£10 million** over a **20-year operational lifetime**



Robustness to changes

Direct values are
easily changed

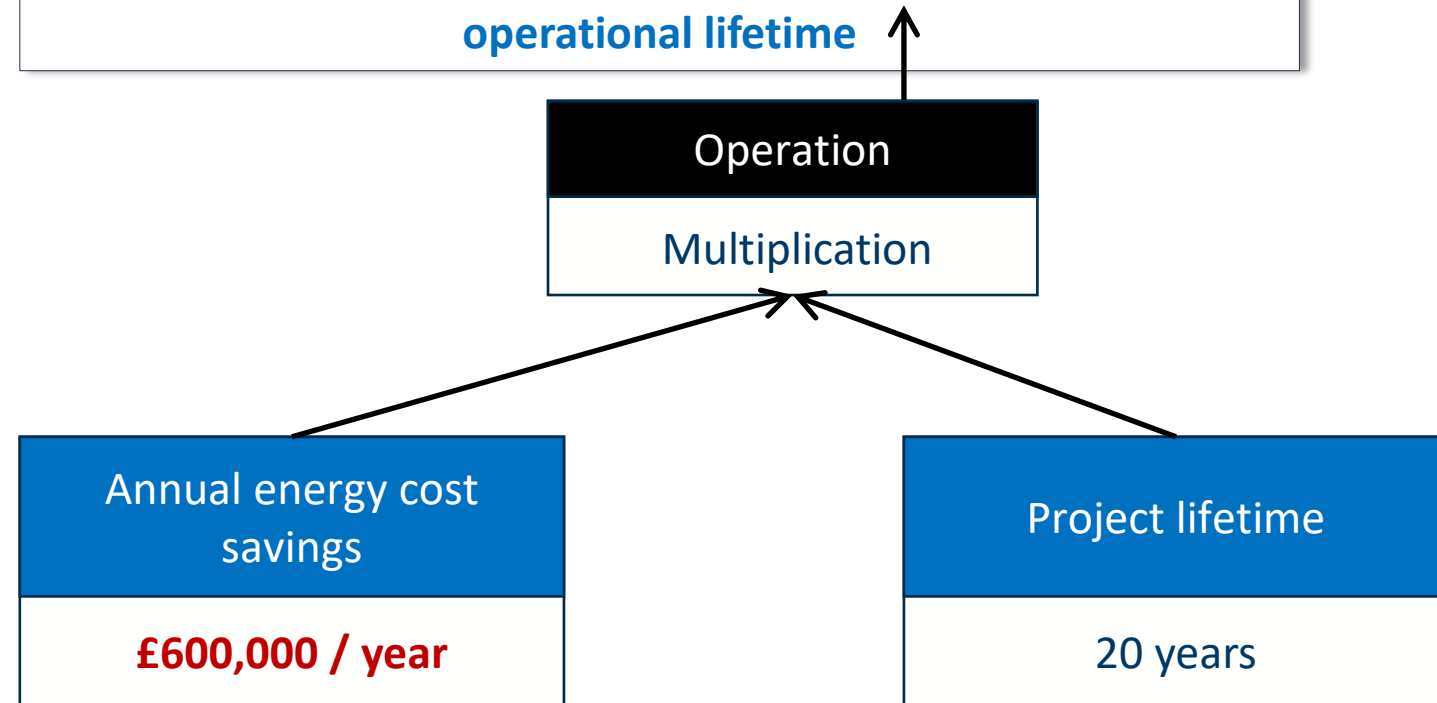
Financial analysis indicates that the programme will deliver approximately **£600,000 in annual energy cost savings**, generating cumulative savings of **£10 million** over a **20-year operational lifetime**



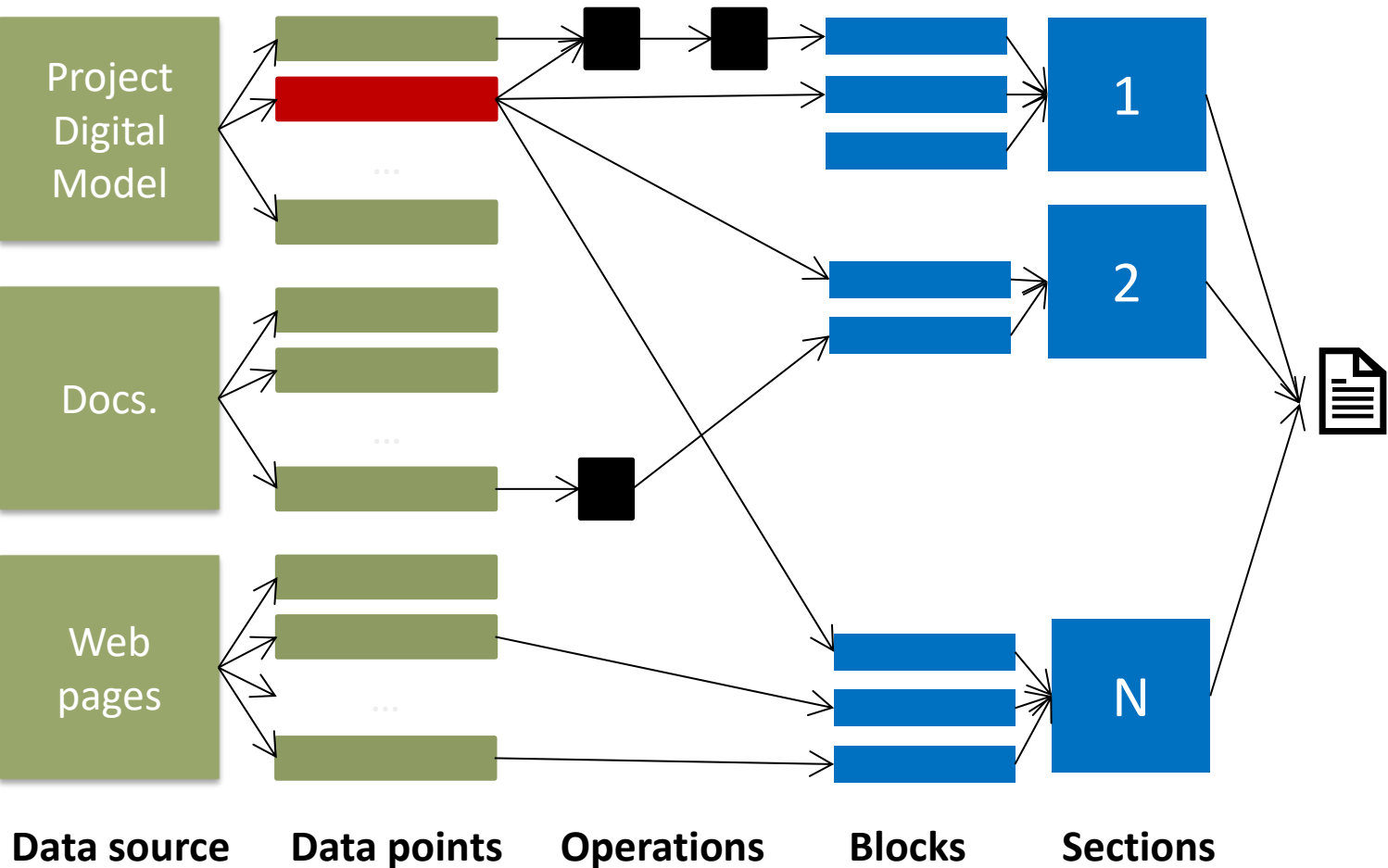
Robustness to changes

We need to just
recompute
operations to fix
other values

Financial analysis indicates that the programme will deliver approximately **£600,000 in annual energy cost savings**, generating cumulative savings of **£12 million** over a **20-year operational lifetime**

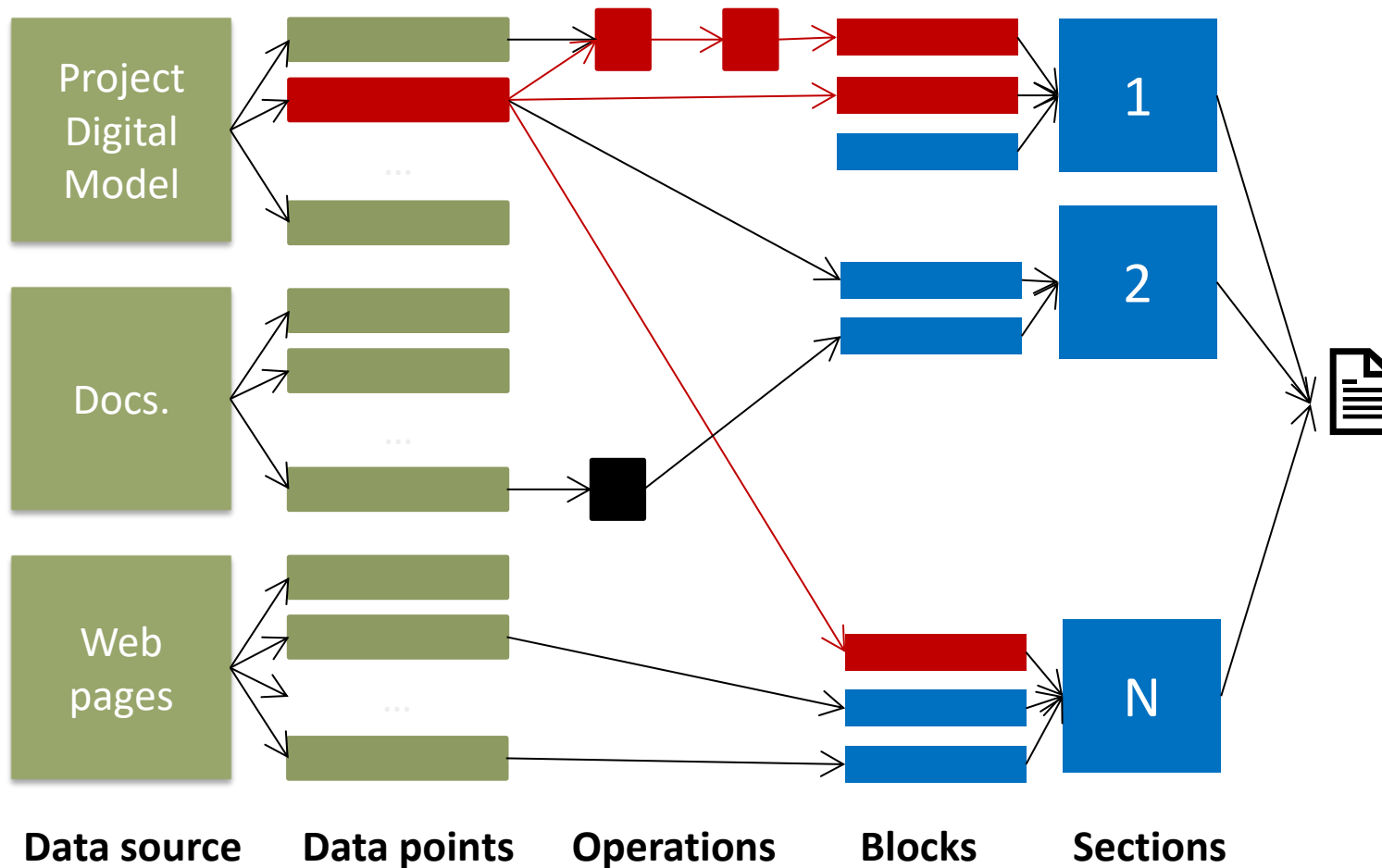


Robustness to changes



If we change a value...

Robustness to changes



If we change a value...

Traversing the graph ahead indicates what elements of the report we need to verify

Conclusions

- Writing bankability reports is a complex task
- But AI can help
- We need with the following capabilities
 1. Manages complex information about a project.
 2. Data in the report can be traced to its origin.
 3. Operations with the data can be verified.
 4. Robust to changes in data
 5. Can help writing some sections of the report
 6. Ideally, by providing data once we can write multiple reports.

Conclusions

- We have built a tool with these capacities



- In the next session, we will show a demo of the tool

Questions?



Dr. Javier Sanz-Cruzado

✉ javier.sanz-cruzadopuig@glasgow.ac.uk

